Neural Holodeck: Real-Time Sim-to-Real Environment Generation via Seamless Latent Diffusion

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Abstract

Creating realistic 3D environments is constrained by the authoring bottleneck of complex modeling and texturing, while generative AI lacks precise spatial control. To address this, we introduce the **Neural Holodeck**, a system that bridges the domain gap between geometric simulation and photorealistic rendering [1]. Our approach uses simple low-poly blockouts with semantic labels (ADE20K) in Unity to guide a **Stable Diffusion XL (SDXL)** model via **ControlNet Union** [2]. A key innovation is the implementation of a **Circular Padding** mechanism within the UNet architecture, which forces the model to treat image boundaries as continuous neighbors, eliminating seam artifacts in 360° generation [3]. We demonstrate that our system generates high-fidelity, seamless 2K/4K panoramas on consumer hardware (RTX 5070) in interactive timeframes (3–5 seconds), projecting the result back into a hybrid physics environment for immediate user immersion.

1 Introduction

Creating immersive 3D environments traditionally suffers from a significant authoring bottleneck. The manual processes of UV mapping, PBR texturing, and baking lighting are labor-intensive. While text-to-image models have advanced rapidly, they lack precise spatial control [1]. Users cannot easily dictate the placement of objects or ensure geometric consistency through text prompts alone.

We introduce the **Neural Holodeck**, a pipeline that treats a Game Engine (Unity) as a semantic signal generator for a Latent Diffusion Model (LDM). By using "Semantic Blocks"—simple primitives colored to match the ADE20K dataset—we provide the deterministic structure lacking in pure AI generation [4].

Our core technical contributions are:

1. A **One-Shot Texture Baking** pipeline that captures a 360° semantic scene from Unity, processes it via SDXL ControlNet, and projects it back onto an inverted sphere.
2. A **Seamless Tiling Algorithm** that modifies the padding mode of convolutional layers in the diffusion model, solving the "seam line" artifact inherent in panoramas [3].
3. A **Hybrid Physics** implementation where visual immersion is provided by the AI hallucination, but physical collision is handled by the original low-poly geometry.

2 Methods

This section details the Neural Holodeck pipeline, comprising Semantic Capture, Neural Synthesis, and Immersive Projection.

2.1 Semantic Capture (Unity)

We employ a custom `SegmentationCamera` positioned at the center of the environment to capture the user's blockout. The camera renders the scene into a Cubemap, which is immediately converted into an Equirectangular Projection (2048×1024).

The mathematical mapping from 3D Cartesian coordinates to 2D image coordinates (u, v) is:

$$\begin{aligned} \phi &= \arctan(z/x) \\ \theta &= \arctan(y/\sqrt{x^2+z^2}) \\ u &= \frac{\phi}{2\pi} + 0.5, \quad v = \frac{1}{\pi}\arcsin(y/R) + 0.5 \end{aligned} \tag{1}$$

2.2 Neural Synthesis (Python)

The Python server receives the semantic map and uses **ControlNet Union** [2] to condition **Stable Diffusion XL**.

2.2.1 Seamless Tiling via Circular Padding

Standard CNNs use zero-padding, which creates a visible vertical line where the panorama wraps. We propose a dynamic architecture modification (Algorithm 1) that forces the model to treat image boundaries as connected neighbors.

Algorithm 1: Seamless Circular Padding Injection

Require: Pre-trained Diffusion Model M (UNet & VAE)
Ensure: Model with seamless boundary handling
procedure MakeSeamless(M)
  for each module L in M.modules **do**
    if L is Conv2d **then**
      if L.padding is $(1, 1)$ **then**
        L.padding_mode ← `'circular'`
      end if
    end if
  end for
end procedure
Call MakeSeamless(M.unet)
Call MakeSeamless(M.vae)

Mathematically, this alters the convolution operation to:

$$(f * g)[n] = \sum_{m=-M}^{M} f[(n-m) \pmod N] g[m] \tag{2}$$

2.3 Immersive Projection (Unity)

The generated image is projected back into the 3D world. To maintain visual fidelity at the "poles" (zenith/nadir), we employ a specific texture import strategy (Algorithm 2).

Algorithm 2: Runtime Texture Import Strategy

Require: Raw Byte Stream B from Server
Ensure: High-Fidelity Spherical Texture T
procedure ProcessTexture(B)
  $T \leftarrow$ LoadImage(B)
  T.FilterMode ← `Trilinear` *// Smooths aliasing*
  T.AnisoLevel ← 16 *// Fixes oblique blur at poles*
  T.WrapMode ← `Repeat` *// Ensures seamless wrapping*
  Apply(T)
end procedure

We generate a procedural sphere with inverted normals, ensuring the texture is rendered on the *inside* face. The user views this sphere but collides with the original invisible blockout geometry.

3 Results & Analysis

We evaluated the Neural Holodeck on an NVIDIA RTX 5070 GPU. The generation time for a 2K panorama averages **3–5 seconds**.

3.1 Visual Fidelity

The system successfully translates low-poly blockouts into high-frequency textures while respecting the original perspective. Figure 1 demonstrates the effectiveness of the circular padding, showing zero seam artifacts at the edges.

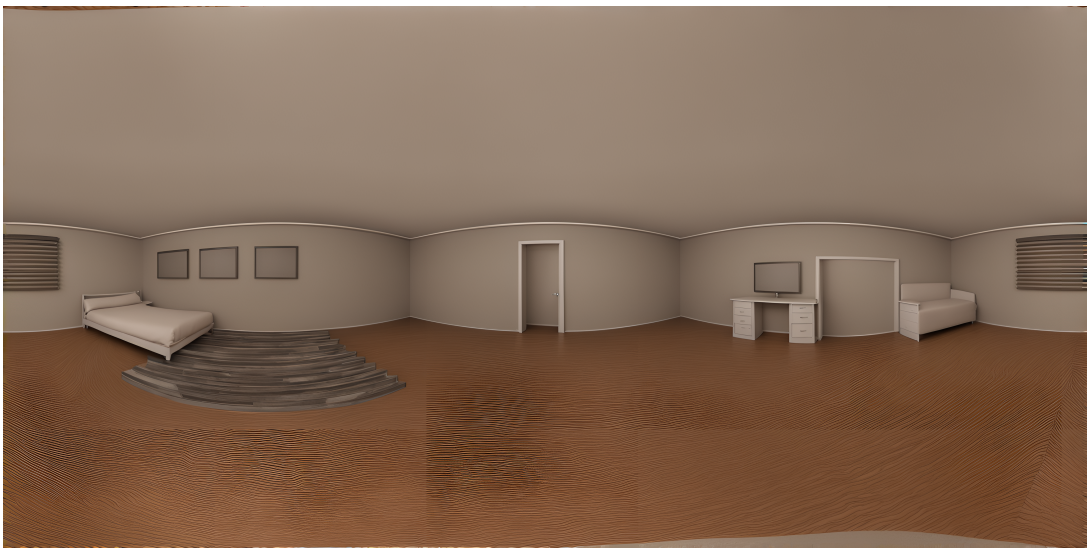


Figure 1: The raw 360° Equirectangular output (2048×1024). The left and right edges match perfectly due to Algorithm 1.

3.2 Immersive Consistency

Figure 2 demonstrates the in-engine experience. The semantic consistency provided by ControlNet ensures that if the user places a pink block for a "bed," the AI generates a realistic bed in that exact location.



Figure 2: In-Engine View. Left: Bedroom configuration. Right: Living room workspace. The AI visuals align perfectly with the invisible Unity colliders.

3.3 Limitations

We observed the "pole" problem, where the zenith and nadir suffer from geometric distortion inherent to equirectangular projection. While anisotropic filtering (Algorithm 2) mitigates the blur, the geometric pinching remains. Additionally, the generated lighting is static and does not respond to dynamic Unity lights.

4 Conclusion

The Neural Holodeck demonstrates that Game Engines can serve as effective control interfaces for Generative AI. By modifying the padding architecture of SDXL, we achieved seamless panoramic generation suitable for real-time immersive applications. This pipeline democratizes 3D environment creation, allowing users to rapidly prototype photorealistic spaces from simple geometric blocks.

References

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